

Chandra Prakash

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RESEARCH INTERESTS

I study quantum fields in nontrivial gravitational backgrounds, ranging from cosmological spacetimes to black-hole geometries. I am particularly interested in **Quantum Field Theory in Curved Spacetime, EFT of Gravity, Asymptotic Symmetries & Holography** and **Early Universe Cosmology**.

EDUCATION

Indian Institute of Technology Guwahati (IITG), Assam, India

- M.Sc. Physics (GPA: 8.65/10) July 2023 – June 2025
 - Thesis: *From Quantum Fields to Cosmic Structures: Tracing the Early Universe through Correlation Functions*
 - Supervisor: Prof. Dr. Debaprasad Maity
 - Focus: Particle production during inflation, correlation functions, cosmological observables.

Savitribai Phule Pune University, Maharashtra, India

- B.Sc. Physics (GPA: 9.38/10) June 2020 – June 2023
 - Thesis: *Non Commutative Geometry as quantization of spacetime*
 - Supervisor: Prof. Lalita Rane
 - Graduated among top 5% of class.

PUBLICATIONS & PREPRINTS

- **Chandra Prakash** and Rajesh Karmakar. *Canonical quantization of massive vector field in Schwarzschild black hole background*. Under Review in **Physical Review D. (PRD)**, 2026. [arXiv:2605.27047]
- **Chandra Prakash** and Debaprasad Maity. *Cosmological correlators from the Inflation end to CMB sky via reheating*. Under Review in **Journal of High Energy Physics (JHEP)**, 2026. [arXiv:2603.24635]
- Anchita Chetia, Nirmalya Kajuri, and **Chandra Prakash**. *Momentum Space Correlation Functions in 2D Galilean Conformal Algebra*. Published in **Journal of High Energy Physics (JHEP)**, 2025. [arXiv:2512.21733]

RESEARCH EXPERIENCE

Shanghai University, China

- Dirac Bracket and QFT in curved spacetime 2026 January – Present
 - **Collaborator:** Dr. Rajesh Karmakar
 - **Research Focus:** Quantum field theory in curved spacetime; black hole backgrounds; vacuum structure.
 - **Skills:** Quantum field theory in curved spacetime; hidden symmetries; black hole physics, constrained hamiltonian dynamics.
 - **Methodology:** Using Analytical methods to study field dynamics in curved spacetime, including basis construction via Löwdin orthogonalization and constrained systems analysis using Dirac-type methods.

Indian Institute of Technology Mandi, Himachal Pradesh, India

- Non-relativistic limit of AdS/CFT in momentum space Summer 2025 – 2025 December
 - **Guide:** Dr. Nirmalya Kajuri
 - **Research Focus:** Studying Galilean Conformal algebra in momentum space to deduce the constraints on relativistic AdS/CFT limits.
 - **Skills:** Bootstrap techniques, conformal ward identities, Wigner-Innou contraction.
 - **Methodology:** Analyzed Galilean Conformal Algebra (GCA) via Wigner-Inönü contraction; derived generators in position and momentum space. Deduced and solved Ward identities for two- and three-point functions.

Indian Institute of Technology Guwahati, Assam, India

- Cosmological Correlations during Reheating 2024 July – 2025 April
 - **Guide:** Prof. Dr. Debaprasad Maity
 - **Research Focus:** Cosmological collider physics and early-universe particle production.
 - **Skills:** Quantum Field Theory in curved spacetime, Schwinger-Keyldish formalism, cosmological perturbation theory, numerical methods, Python.
 - **Methodology:** Worked on evolving cosmological correlators from the end of inflation to CMB Sky, particle production, and CMB effects.

**TECHNICAL
EXPERTISE**

Physics: Quantum Field Theory in Curved Spacetime; Cosmological Perturbation Theory; Effective Field Theory and Schwinger-Keldish formalism; Constrained Hamiltonian Systems; Black Hole Quantum Physics; Conformal Symmetry and Ward Identities.

Computational: Python, C++, Fortran, Mathematica.

**CONFERENCES &
SCHOOLS**

- Loop the Loop Workshop (2024): Self-force and loop corrections in GR.
- GIAN Course on Standard Model EFT (2024).
- IUCSS Summer School (2021): Lorentz/CPT violation.
- String 2021 Conference.

OUTREACH

- Gave a Zoom talk on arXiv:2603.24635 at Swansea Cosmology Group.
- Maintains detailed **Conformal Field Theory**, **String Theory** and **Cosmology** notes at physicsphile.github.io.
- Presented Black Hole Entropy work at Zonal Level Aavishkar research event.
- First prize in poster competition (Pune University).

REFERENCES

- **Dr. Nirmalya Kajuri**
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Indian Institute of Technology Mandi
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- **Prof. Debaprasad Maity**
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- **Prof. Ashwini Kumar Sharma**
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